

COOPER MCALLISTER

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Current senior seeking full-time engineering roles in cutting-edge aerospace and defense technology. Two years of experience in the aviation and defense industry, and hands-on project experience in PCB and electrical design.

EDUCATION

LETOURNEAU UNIVERSITY, ELECTRICAL AND COMPUTER ENGINEERING,
MECHATRONICS CONCENTRATION, B.S. CLASS OF 2027 | Honors College | President's List | GPA 4.0

WORK EXPERIENCE

ELECTRICAL ENGINEERING INTERN, SIERRA NEVADA CORPORATION

MAY 2026 – AUGUST 2026

- Created a VBA-based system used to record over 60 recurring non-conformances, saving time in production
- Led a meeting with 15 engineers and technicians to implement the system in the existing non-conformance process
- Worked as a liaison engineer in a hanger to resolve issues on C-130 aircraft modification installations
- Used AutoCAD and 3DEXperience PDM to revise 23 harness, wire diagram, and installation drawings for C-17 mods

AVIONICS AND ELECTRICAL SYSTEMS ENGINEERING INTERN, TEXTRON AVIATION

JUNE 2025 – AUGUST 2025

- Wrote Python scripts to automate Garmin G1000 avionics testing for Cessna aircraft, decreasing testing time by 80%
- Gained experience with Siemens Capital (Mentor Graphics), MS Office, VS Code, and Textron software tools
- Assisted with design changes due to part obsolescence in the context of FAA certification

ELECTRIC CIRCUITS SUPPLEMENTAL INSTRUCTOR, LETOURNEAU UNIVERSITY

JANUARY 2026 – PRESENT

- Lead weekly review and practice sessions to improve students' understanding of difficult topics
- Created over 80 pages of notes and practice problems using LaTeX to use as teaching material
- Peer tutor for 2 semesters for electric circuits, calculus, and differential equations

PROJECT EXPERIENCE

AVIONICS TEAM LEAD, LETOURNEAU UNIVERSITY SPACE PROGRAM

AUGUST 2025 – PRESENT (LEAD), AUGUST 2024 – AUGUST 2025 (MEMBER)

- Manage development of avionics systems with live telemetry for high-power amateur rocketry
- Created a new remote ignition system for testing experimental rocket motors (over 80 hours)
- Gained skills in PCB design using KiCad, circuit design, and embedded programming in C for the Raspberry Pi

ENGINEERING DESIGN, PROJECT MANAGEMENT - AUTONOMOUS VEHICULAR INTEGRATED DEMINING (AVID)– 100+ HOURS

JANUARY 2026 – MAY 2026

- Singlehandedly created the entire electrical system, including a hand-soldered board, with a \$100 budget
- Developed system architecture using SysML and implemented hardware design in KiCad
- Fabricated circuit board with ESP32 and stepper drivers, wire harnesses, and metal detector circuit

VHDL AND FPGA DEVELOPMENT - TIC TAC TOE (TEAM LEAD, PROGRAMMER) – 25+ INDIVIDUAL HOURS

APRIL 2025 – MAY 2025

- Wrote code used in the Tic Tac Toe Game in VHDL for an AMD Artix 7 FPGA using the AMD Vivado Design Suite
- Led a team of four to manage requirements, wrote an IEEE format project report, and gave an in-class presentation

SKILLS

- Microsoft Office Suite: Word, Excel, PowerPoint, SharePoint
- CAD: SolidWorks, Mentor Graphics, AutoCAD, KiCad EDA, 3DEXperience / ENOVIA / PLM
- Programming: Embedded C, C++, Python, MATLAB, GitHub, LabVIEW, VHDL, Visual Studio Code
- Electronics: Oscilloscopes and test equipment, analog and digital circuit design, PCB design and bring-up

- Aviation: Familiarity with SAE AS50881, TO 1-1A-14, aircraft production and QA processes, 20 hours flight instruction